

AKSHAR VADWALA

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OBJECTIVE

Systems and data enthusiast seeking a full-time role in backend development or data engineering. Adept at designing scalable architectures using Python, Java, cloud platforms, and distributed computing frameworks to drive impactful technical solutions.

EDUCATION

MS In Computer Science, The University of Texas at Dallas

Aug 2025 - May 2027 | GPA: 3.88

BE In Computer Engineering, MBIT, CVM University

Sep 2021 - May 2025 | GPA: 3.94

EXPERIENCE

Software Engineer Intern, Smart Data Solutions, Dallas, Texas

May 2026 - Present

- Improved healthcare claim data extraction pipelines processing CMS-1500, UB-04, and Dental forms by leveraging OCR() and Vision Language Models, keeping OCR accuracy above 99.5% and maximizing automation over 90%.
- Troubleshoot and optimized Java and SQL-based production systems in linux environments, resolving client specific workflow failures and enhancing performance, reliability, and SLA compliance.
- Supported migration of enterprise data assets from on-premises infrastructure to AWS and developed recommendation models using matrix factorization and cosine similarity for client expansion analysis.

Data Engineer Intern, Minix Fintech (HFT), Gujarat, India

Jan 2025 - May 2025

- Designed and optimized high throughput data pipelines (equity, crypto, commodities) using Python, Prefect, Bash and ClickHouse to automate the daily ingestion of 500+ GB of tick-by-tick data, improving throughput & latency by 60%.
- Built ETL workflows for real-time data (BSE, MCX, Binance) into Azure Blob, enabling low-latency trading analytics.
- Worked in Linux based environments analyzing performance bottlenecks. Implemented expiry logic for F&O instruments and enhanced pipeline reliability by upgrading Prefect (v2.0 → v3.0), with version control via Bitbucket and Jira.

PROJECTS

Multi-Object Tracking Analytics (Python, OpenCV, Kalman Filter, DeepSORT)

Developed a scalable tracking by detection pipeline for trajectory analytics in high density environments using YOLOv8, preserving unique identities across occlusions via DeepSORT (ResNet18) and Kalman Filter temporal motion prediction.

Real-Time Named Entity Recognition on Reddit Stream (Python, Spark, Kafka, ELK)

Engineered a real-time data streaming pipeline to extract named entities from Reddit comments using PySpark and NLTK. Integrated Apache Kafka for message streaming and Kibana through ELK stack for visualizing top entities over time intervals.

Big Data Processing using Spark (Python, Spark, NLTK)

Implemented distributed Named Entity extraction and WordCount using PySpark + NLTK, processing large text corpora. Built a scalable movie-plot search engine with Spark RDDs using TF-IDF and cosine similarity for multi-term query ranking.

Career Recommendations using ML for Matriculants (Python, Scikit-learn, MySQL)

Engineered a random forest regression machine learning model for personalized career suggestions, supported by Python, Flask, and MySQL. Created computer adaptive test interface using HTML, CSS, and JavaScript to recommend career streams.

IoT based Weather App (Java, Android, AWS, SQLite)

Built an Android app using Java that integrates with IoT sensors (DHT11, BMP180, water sensor) to fetch real-time data from surroundings. Implemented MQTT protocol to transmit sensor data to AWS IoT Core.

TECHNICAL KNOWLEDGE

- **Programming Languages:** Python, Java, C, C++, JS
- **Backend Dev & Web:** Flask, Django, Servlets, JSP, JSTL, HTML5, CSS3, PHP, XML, Bootstrap, Wordpress
- **Libraries & Frameworks:** Prefect, Scikit-learn, JDBC, FastAPI, Hadoop, Pandas, NLTK, Spark, Kafka, GraphX
- **Database:** MySQL, Postgres, Clickhouse, MongoDB
- **Platforms & OS:** AWS (S3, EC2, EMR, IoT Core), Azure (Blob storage, DataLake, DataFactory), Linux, Windows
- **Development Tools:** VS code, IntelliJ, Eclipse, Android Studio, Github, Gitlab, Bitbucket, Jira

PUBLICATIONS

Career Recommendation Using Machine Learning for Secondary Education. IEEE IATMSI Conference, 2025, Gwalior, India.

Network Based Smart Attendance System for Students. 2024 International Research Journal of Innovations in Engineering and Technology - IRJIET, 8(8), 287-291.

A Comprehensive Review Of Digitalized Attendance Systems For Various Institutions. 2024 International Journal of Engineering Applied Sciences & Technology (IJEAST), 2024, 9(3), 119-123.

EXTRACURRICULAR ACTIVITIES

Testing Center Proctor, UT Dallas (Jan 2026 - May 2026): Administer secure exams, manage Windows systems, enforce academic protocols, and provide service oriented student support.

Chairperson, IEEE CS MBIT SB (Oct 2022 - Oct 2024): Led 15-member committee for 10+ technical events, overseeing logistics, promotions, and guest engagement. Tutored MBIT placement cell students in Java and C.